# CSE 6730: Modeling and Simulation

Final Project Report

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## Undulatory Locomotion Simulator in Overdamped Environments

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**GitHub Repository:** https://github.com/jianflin/CSE6730_GM_Modeling


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**Abstract**

Undulatory locomotion is a strategy used by a wide range of biological and robotic systems, from microscopic swimmers to limbless and multi-legged animals. In overdamped environments, where dissipative forces dominate inertia, motion is generated entirely by cyclic body deformations interacting with the environment. This project develops a physics-based simulator for two such systems: a limbless undulatory body in a viscous fluid, and a centipede-like multi-legged body interacting with a Coulomb friction substrate. The modeling approach combines a reduced-order rigid-link representation of the body with resistive force theory (RFT) for the local environmental interaction, and is implemented as a modular Python package. At each time step, the simulator prescribes a body wave (and, for the centipede, leg shoulder oscillations and a stance/swing contact pattern), computes local resistive forces, and solves the force-free condition for the body twist. Parameter sweeps over body wave amplitude and body–leg phase lag reproduce expected qualitative behaviors: isotropic viscous drag yields negligible net displacement, anisotropic drag enables forward propulsion with a non-monotonic dependence on amplitude, and centipede performance exhibits a sharp peak in amplitude–phase space near a body–leg phase lag of 180°. The simulator provides an interpretable, modular framework for comparing limbless and legged undulatory locomotion under common assumptions.

# 1 Project Description

**The real-world systems.** Many animals move using *undulatory locomotion*: they propagate a traveling wave of body bending along their body, and the interaction between this wave and the surrounding environment produces net forward motion. Snakes slithering across sand, nematodes swimming in fluid, and centipedes running on the ground are all examples. Their robotic counterparts—snake-like robots and multi-legged crawlers—use the same basic principle: cyclic shape changes that, through environmental interaction, are converted into displacement of the body. Figure 1 shows three of these systems and their abstraction into a chain of rigid links connected by joints.

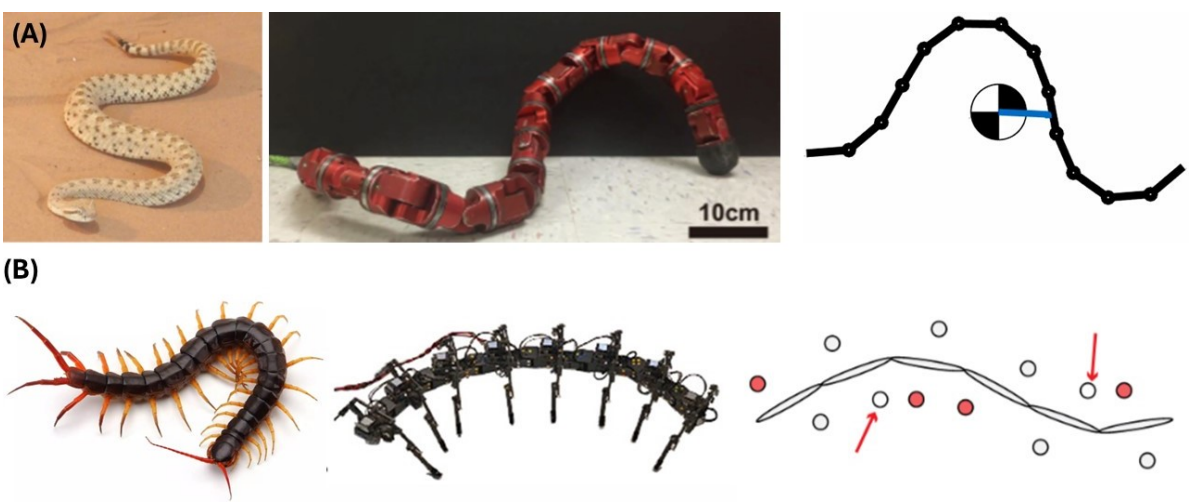


Figure 1: Undulatory locomotion across biological and robotic systems and its abstraction into a reduced-order model. (A) Limbless systems: a sidewinder snake, a snake robot, and the rigid-link representation used in this project. (B) Multi-legged systems: a centipede, a centipede-like robot, and the corresponding leg-and-body schematic. Both classes of systems are modeled as series-connected rigid links interacting locally with their environment.

**Why overdamped locomotion is interesting.** A defining feature of these systems is that they operate in *overdamped* regimes. Inertia is small compared to dissipative forces (viscous drag, granular resistance, or Coulomb friction). If the body stops deforming, motion stops almost immediately. Net displacement therefore does not come from accelerating mass, but rather from the *geometry* of the cyclic shape change interacting with the environment. This is what makes the problem mathematically interesting: locomotion becomes a question of how a closed loop in shape space maps to a net translation/rotation in physical space.

**Two target systems.** We focus on two representative systems:

- *Limbless undulatory locomotor in a viscous fluid.* The body is a chain of rigid links, and each link experiences anisotropic linear drag with different drag coefficients parallel ($\gamma_x$) and perpendicular ($\gamma_y$) to the link. This model captures the essential physics of microswimmers and snake-like swimming.
- *Centipede-like multi-legged locomotor on a Coulomb friction substrate.* In addition to body undulation, the body carries leg pairs whose shoulder angles oscillate and whose contact state (stance/swing) is coordinated with the body wave through a body–leg phase lag.

**Goal of the project.** The goal is *not* to perfectly replicate the locomotion of any specific animal or robot. Instead, we build a modular, interpretable simulator that captures the dominant mechanics of overdamped undulatory locomotion in a unified framework, and we use this simulator to study how gait parameters (body amplitude and body–leg phase) affect locomotion performance in both limbless and legged systems.

## 2 Literature Review

Geometric mechanics provides a powerful framework for analyzing locomotion in dissipative environments, where motion arises from cyclic body deformation rather than inertia. A central result of this framework is that, under overdamped assumptions, body velocity can be written as a linear mapping from shape velocity, and net displacement over a gait cycle can be predicted from a geometric phase. Hatton and Choset [5] formalized this for general locomotor systems, introducing the *local connection* that relates shape space and position space and showing how Stokes' theorem can be used to evaluate displacement as a height-function area integral.

Building on this framework, Ramasamy and Hatton [8] investigated the geometry of optimal gaits in drag-dominated kinematic systems. Their analysis shows that locomotion performance can be understood and optimized by examining the structure of the local connection and the associated geometric phase, providing a principled way to identify effective deformation patterns rather than relying solely on numerical search.

Recent work has extended geometric mechanics to biological systems and robophysical experiments. Rieser et al. [9] demonstrated that undulatory locomotion across a wide range of organisms—including nematodes, lizards, and snakes—can be described within a low-dimensional shape space, and that these systems often operate near optimal gaits predicted by geometric phase analysis. This suggests a unifying principle that crosses scales and morphologies. In robotics, Chong et al. [2] showed that modulating body wave frequency can significantly improve sidewinding robot performance, illustrating how geometric insights can translate directly into control strategies.

A more recent line of work explicitly addresses multi-legged locomotors. Chong et al. [3] introduced a general control framework for multi-legged systems that combines a body undulation wave with a leg stepping wave parametrized by a duty factor and an effective stride length, drawing on

Hildebrand's classical gait description. This framework treats the body and the legs as two coupled traveling waves and is directly relevant to the centipede model studied here: it provides the parametrization for leg shoulder oscillations and the body–leg phase lag that we adopt in Section 4. Earlier biological work also connects undulatory body motion to interactions with non-fluid media: Maladen et al. [7] studied subsurface "swimming" of the sandfish lizard in granular substrates and showed that an undulatory wave at the appropriate amplitude and wavelength can generate effective propulsion through grains, motivating the use of RFT-style local force models beyond simple viscous fluids.

In parallel with geometric mechanics, *resistive force theory* (RFT) has been widely used to model environmental interactions in viscous fluids and granular media. Originally developed by Gray and Hancock [4] for low-Reynolds-number flagellar swimming, RFT assumes that the force on each body segment depends only on its local velocity and orientation, with separate drag coefficients for motion parallel and perpendicular to the segment. Li, Zhang, and Goldman [6] extended RFT to legged locomotion on granular media ("terradynamics"), demonstrating that empirically fit local force laws—the BB-style empirical force model used in this work is in the same family—can accurately predict whole-body motion of legged robots and animals on sand.

Together, these works support a tractable modeling workflow: RFT supplies a local force model, and a quasi-static force balance provides the global mapping from shape change to body displacement. Previous biological and robotic studies [9, 2, 3, 1] motivate two specific lines of analysis pursued here: (i) the use of low-dimensional, parametrized gaits (amplitude, spatial wave number, frequency, and—for legged systems—duty factor and body–leg phase lag), and (ii) parameter sweeps that compare locomotion performance across body amplitudes and body–leg coordination patterns. The simulator developed in this project combines these ideas into a single framework that can compare limbless and legged undulatory locomotion under common assumptions.

## 3 Conceptual Model

We build a reduced-order conceptual model that captures the essential physics of undulatory locomotion while remaining computationally tractable.

**Body as a chain of rigid links.** The body is represented as a chain of $N$ rigid links connected by $N-1$ revolute joints. The body shape is fully described by the joint-angle vector $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{N-1})$. The position and orientation of the body in the plane is described by $g = (x, y, \theta)$.

**Undulatory gait as a traveling wave.** The gait is a prescribed traveling wave of joint angles:

$$\alpha_{\text{body}}(t, i) = A_{\text{body}} \sin\left(2\pi S_n \frac{i}{N} - 2\pi f t\right), \tag{1}$$

where $A_{\text{body}}$ is the wave amplitude, $f$ is the temporal frequency, $S_n$ is the spatial wave number, and $i$ is the joint index. This single equation captures the full body wave with four parameters: amplitude, frequency, wave number, and phase. Equivalently, the wave can be decomposed into two spatial modes with time-varying weights $w_1(t), w_2(t)$, giving a low-dimensional shape-space representation.

**Centipede leg model.** For the centipede system, the body wave from Eq. (1) is augmented with leg shoulder oscillations on each side of the body and a stance/swing contact schedule. Two features matter:

- Each leg pair has its own shoulder-angle oscillation relative to a rest pose.

- Each leg has a binary contact state (stance vs. swing) that advances along the body in coordination with the body wave, parametrized by a *body–leg phase lag* $\phi_{\text{lag}}$.

Phase lag is the key coordination parameter for centipede locomotion, because it determines whether legs push against the ground when the local body bending is in a configuration that converts those forces into forward thrust.

**Mapping conceptual features to physical phenomena.** The conceptual model maps cleanly onto observable phenomena:

- Body wave $\leftrightarrow$ undulatory bending.
- Viscous anisotropy $\leftrightarrow$ different drag parallel and perpendicular to the body.
- Coulomb friction $\leftrightarrow$ frictional contact with the ground.
- Leg contact pattern $\leftrightarrow$ stance/swing sequence in centipede locomotion.

**Simplifying assumptions.** We assume planar motion, overdamped dynamics (no inertia), and locality of environmental interactions (forces on each link/leg are independent). These assumptions retain the dominant mechanics while enabling a tractable numerical implementation.

# 4 Mathematical Model

The mathematical model maps prescribed shape changes to body motion through three ingredients: link kinematics, a local resistive force model, and a global force/torque balance. We first describe the body kinematics, then the leg gait used for the centipede system, and finally the assembly of the global force balance.

## 4.1 Configuration and Kinematics

The system configuration consists of the body pose and shape variables:

$$g = (x, y, \theta) \in SE(2), \qquad \boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{N-1}). \tag{2}$$

The spatial velocity of link $i$ can be expressed as a linear combination of the body twist $\dot{g}$ and the shape velocity $\dot{\boldsymbol{\alpha}}$:

$$\dot{g}_i \;=\; T_i(\boldsymbol{\alpha})\,\dot{g} \;+\; J_i(\boldsymbol{\alpha})\,\dot{\boldsymbol{\alpha}}, \tag{3}$$

where $T_i$ is the SE(2) transformation that maps body velocity into the frame of link $i$, and $J_i$ is the spatial Jacobian relating joint motion to link motion.

## 4.2 Body Wave and Leg Stepping Wave

The body shape is driven by the prescribed traveling wave of joint angles introduced in Eq. (1):

$$\alpha_{\text{body}}(t, i) = A_{\text{body}} \sin\left(2\pi S_n \frac{i}{N} - 2\pi f t\right). \tag{4}$$

For the centipede system, we additionally describe a *leg stepping wave* together with a contact modulation rule. We follow the formulation of Chong et al. [3], which uses Hildebrand's classical gait description in terms of a duty factor $D \in (0, 1)$ (the fraction of one cycle that a leg spends in stance) and an effective stride length. Each pair of legs at body location $i$ is controlled through a

left and right shoulder angle $\beta_L^i$ and $\beta_R^i$. For the first leg pair, the shoulder angle as a function of the leg phase $\phi_{\text{leg}}$ is

$$\beta_L^1(\phi_{\text{leg}}, 1) = \begin{cases} A_{\text{leg}} \cos\left(\dfrac{\phi_{\text{leg}}}{2D}\right), & 0 \le \phi_{\text{leg}} < 2\pi D, \\ A_{\text{leg}} \cos\left(\dfrac{\phi_{\text{leg}} + 2\pi(1-2D)}{2(1-D)}\right), & 2\pi D \le \phi_{\text{leg}} < 2\pi. \end{cases} \tag{5}$$

The two pieces of Eq. (5) correspond to the stance and swing phases of the leg: during stance ($0 \le \phi_{\text{leg}} < 2\pi D$) the shoulder sweeps backward at one rate, and during swing ($2\pi D \le \phi_{\text{leg}} < 2\pi$) it returns at a faster rate that compresses the remaining $(1-D)$ fraction of the cycle. The amplitude $A_{\text{leg}}$ controls the angular sweep of each shoulder. The right-side leg $\beta_R^i$ is offset from $\beta_L^i$ by a fixed half-cycle phase, and the leg phase along the body is offset from leg pair to leg pair so that the leg stepping wave travels along the body.

The body wave and the leg stepping wave are coupled through a single coordination parameter, the *body–leg phase lag* $\phi_{\text{lag}}$, defined as the phase shift between the local body wave at body location $i$ and the leg phase $\phi_{\text{leg}}$ at the same location. Together with the contact modulation (stance when the leg is in the first piece of Eq. (5), swing otherwise), $\phi_{\text{lag}}$ fully determines whether each leg is pushing against the ground when the local body bending is in a configuration that converts that push into forward thrust.

### 4.3 Local Force Model: Resistive Force Theory

The resistive force on each link is a function of its local velocity:

$$\mathbf{F}_i \;=\; \mathbf{D}_i(\dot{g}_i), \tag{6}$$

where $\mathbf{D}_i$ is the local resistive law:

- **Viscous drag (linear, anisotropic):** $\mathbf{F}_i = -\gamma_x v_\parallel \hat{\mathbf{e}}_\parallel - \gamma_y v_\perp \hat{\mathbf{e}}_\perp$, where $v_\parallel, v_\perp$ are the components of the link velocity along and perpendicular to its axis.
- **Coulomb friction (anisotropic, sign-dependent):** $\mathbf{F}_i = -\mu_x \operatorname{sgn}(v_\parallel)\,\hat{\mathbf{e}}_\parallel - \mu_y \operatorname{sgn}(v_\perp)\,\hat{\mathbf{e}}_\perp$, applied only when the segment is in contact with the ground (for legged systems, this corresponds to the stance phase).
- **BB empirical model:** a piecewise/empirical force law fit from granular drag measurements, in the family of the terradynamics force laws of Li et al. [6] and the granular RFT used in the sandfish swimming studies [7].

### 4.4 Global Force Balance and Force-Free Condition

For the centipede, the same approach is applied to legs in stance, which contribute additional contact forces $\mathbf{F}_j^{\text{leg}}$. The total wrench on the body, summed over all links and active legs, is a function of the body twist $\dot{g}$ and the prescribed shape velocities:

$$W_{\text{net}}(\dot{g}; \boldsymbol{\alpha}, \dot{\boldsymbol{\alpha}}) \;=\; \sum_i \begin{bmatrix} \mathbf{F}_i \\ \mathbf{r}_i \times \mathbf{F}_i \end{bmatrix} + \sum_{j \in \text{stance}} \begin{bmatrix} \mathbf{F}_j^{\text{leg}} \\ \mathbf{r}_j \times \mathbf{F}_j^{\text{leg}} \end{bmatrix}. \tag{7}$$

Under the overdamped assumption, the net wrench must vanish:

$$\boxed{W_{\text{net}}(\dot{g};\, \boldsymbol{\alpha}, \dot{\boldsymbol{\alpha}}) \;=\; 0.} \tag{8}$$

Equation (8) is the central equation of the simulator. At every time step, the body twist $\dot{g}$ is determined as the solution that makes the total force and torque zero, given the prescribed shape and shape velocity.

### 4.5 Geometric Mechanics Interpretation

When the local force law is linear (e.g., viscous drag), Eq. (8) yields a linear relation

$$\dot{g} \;=\; A(\boldsymbol{\alpha})\,\dot{\boldsymbol{\alpha}}, \tag{9}$$

where $A(\boldsymbol{\alpha})$ is the *local connection*, the matrix that maps shape velocity to body velocity. The net displacement over one gait cycle is then a path integral in shape space,

$$\Delta g \;=\; \oint A(\boldsymbol{\alpha})\,d\boldsymbol{\alpha}, \tag{10}$$

which by Stokes' theorem can be approximated as the area integral of the curl of $A$ (the *height function*). For the Coulomb friction case, the local force law is sign-dependent and therefore *not* strictly linear in the body twist, so the local connection is not available in closed form. In practice, however, the linear local-connection picture is often *assumed* to hold approximately, and the corresponding height function is recovered *empirically* from simulation: by sweeping shape loops and recording the resulting body displacement, an approximate height function can be constructed that captures the same conceptual relationship between closed loops in shape space and net body displacement. We do not directly construct this empirical height function in this report, but we rely on the same conceptual picture to interpret the amplitude and phase-lag sweeps in Section 6.

## 5 Simulator Implementation

The simulator is implemented in Python using NumPy and SciPy, organized as a set of focused modules.

### 5.1 Modules

`kinematics.py`
: SE(2) transforms and their inverses, adjoint mappings, forward kinematics for body and leg frames, link positions, link velocities, and spatial Jacobians.

`forces.py`
: Local force models: anisotropic viscous drag, anisotropic Coulomb friction, and the BB empirical force model. Also handles segment-level integration of distributed forces along each link.

`solver.py`
: Assembles the total body wrench from all links and active legs and solves the force-free condition (8) for the body twist $\dot{g}$ at each time step using a nonlinear solver.

`simulation.py`
: Prescribes body and leg gaits (joint trajectories, leg shoulder oscillations, contact schedules), drives the solver, integrates the body pose forward in time, and stores trajectory and state history.

`test_kinematics.py`, `test_simulation.py`
: Unit and end-to-end tests verifying transformations, Jacobians, local velocities, and the full simulation pipeline.

### 5.2 Numerical Procedure

At each time step the simulator executes the following steps:

1. Prescribe body and leg gait inputs (joint angles, leg shoulder angles, contact pattern).
2. Compute joint angles and joint velocities from the gait.
3. Compute the local velocity of each link (and each active leg contact) from Eq. (3).
4. Compute local resistive forces from the chosen force model.
5. Sum forces and torques to assemble the total body wrench as a function of the unknown body twist $\dot{g}$.
6. Solve Eq. (8) for $\dot{g}$.
7. Integrate the body pose: $g(t + \Delta t) = g(t) + \dot{g}(t)\,\Delta t$.
8. Compute displacement per cycle and visualize trajectories.

### 5.3 Why This Method Fits the Overdamped System

Because inertia is negligible, the body has no momentum to track across time steps; the only state that needs to be integrated is the body pose $g$. The body twist at every instant is fully determined by the instantaneous shape and shape velocity through the force-free condition. This makes a quasi-static, time-stepping solver natural: each time step solves an algebraic (rather than differential) equation for $\dot{g}$, then advances the pose. The method is modular by construction, since each component (kinematics, force model, solver, gait) can be swapped independently.

### 5.4 Alternatives Considered but Not Used

- *Full inertial multibody dynamics:* would require integrating accelerations and would be wasted effort in the overdamped regime, where inertia is by assumption negligible.
- *CFD or DEM:* would resolve the surrounding fluid or grain medium directly. This is far more accurate but computationally expensive and not well aligned with our goal of a reduced-order, interpretable simulator.
- *Continuous (PDE) body models:* elegant for a snake-like backbone, but harder to integrate with discrete leg contact and less convenient for direct comparison with link-based robotic designs.

The methods above are not chosen because the project specifically targets a reduced-order, modular, interpretable simulator suitable for parameter sweeps.

## 6 Experimental Results and Validation

We use the simulator to study how gait parameters control locomotion performance. The primary performance metric is *displacement per cycle*, measured in body lengths (BL). We run two main experiments: an amplitude sweep and a body–leg phase lag sweep, complemented by trajectory and snapshot diagnostics.

## 6.1 Gait Overview

Figure 2 shows the prescribed gait for the centipede system: the body joint angle wave, the left and right leg shoulder angle deviations from rest, and the leg contact pattern (black indicates stance). Several features are visible: the body joints execute a smooth traveling wave, the left and right legs oscillate with the same period as the body wave but with a phase shift between sides, and the contact pattern advances along the body in a wave-like manner. This validates the input-generation side of the simulator: the body wave, leg oscillations, and stance/swing timing are coordinated consistently and are consistent with the intended centipede gait. The phase relation between the body wave and the leg contact pattern is the key control parameter studied in Section 6.3.

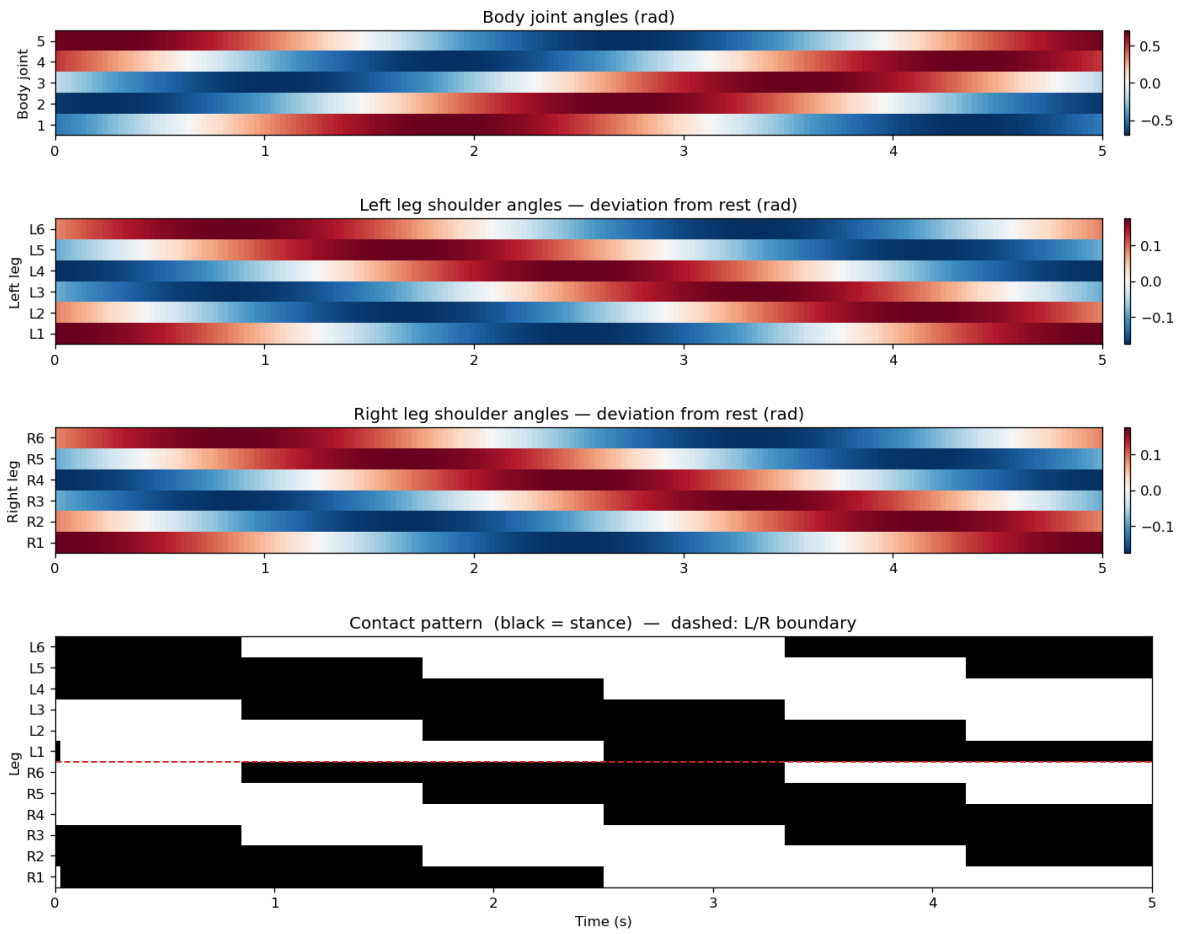


Figure 2: Centipede gait inputs over five seconds. From top to bottom: body joint angles (rad), left leg shoulder angle deviations from rest, right leg shoulder angle deviations, and contact pattern (black = stance). The body wave, leg oscillations, and stance/swing schedule are coordinated consistently with the prescribed body–leg phase lag, validating the input-generation pipeline.

## 6.2 Amplitude Sweep

**Procedure.** For the limbless viscous system, we sweep body wave amplitude $A_{\text{body}}$ from 0° to 80° and compare two fluid environments: isotropic drag ($\gamma_y/\gamma_x = 1$) and anisotropic drag ($\gamma_y/\gamma_x = 5$). For the centipede Coulomb system, we sweep amplitude over the same range with the body–leg

phase lag fixed at the empirically optimal value of 180°. In all cases we report displacement per cycle in body lengths (BL/cycle).

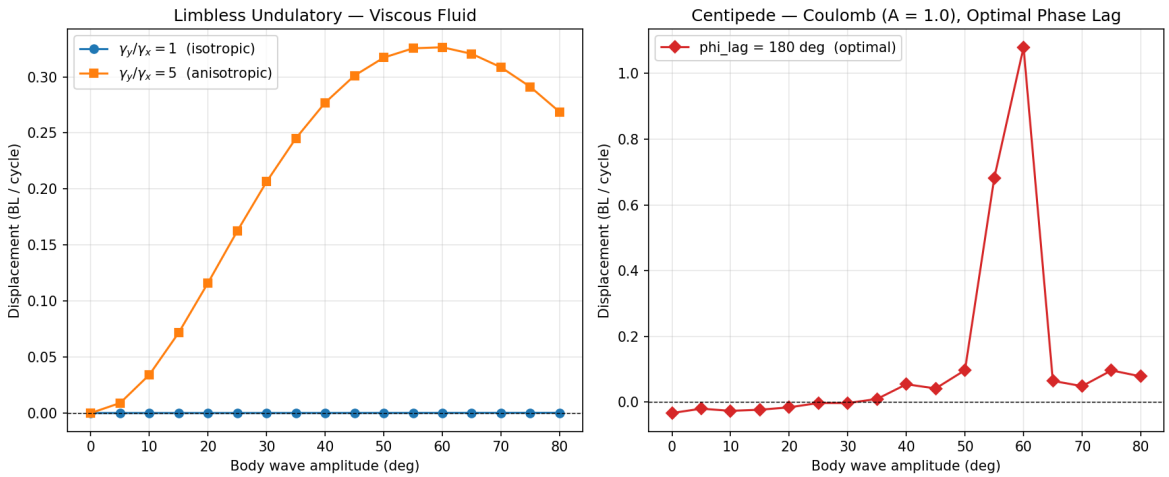


Figure 3: Body amplitude sweeps. *Left:* limbless undulatory locomotor in viscous fluid. Isotropic drag ($\gamma_y/\gamma_x = 1$) yields essentially zero displacement at all amplitudes, while anisotropic drag ($\gamma_y/\gamma_x = 5$) produces forward locomotion that peaks near 55–60°. *Right:* centipede with Coulomb friction at the optimal phase lag, showing a sharp performance peak near 60°. The figure supports two key validation conclusions: environmental anisotropy is necessary for propulsion in the viscous limbless case, and intermediate amplitudes are optimal in both systems.

**Limbless interpretation.** The isotropic drag curve in Fig. 3 (left) lies on zero across the entire amplitude range. This is the expected limiting-case behavior: when $\gamma_y/\gamma_x = 1$, the environment provides no directional preference, so no amount of symmetric body undulation can produce net displacement. With anisotropic drag ($\gamma_y/\gamma_x = 5$), displacement grows nearly monotonically with amplitude up to roughly 55–60° ($\sim 0.32$ BL/cycle), and then decreases. This non-monotonic trend is consistent with the geometric-mechanics intuition that displacement scales with the area enclosed by the gait loop in shape space, but at very large amplitudes the loop deforms in a way that no longer increases the relevant geometric phase.

**Centipede interpretation.** The centipede curve in Fig. 3 (right) shows a *sharp* performance peak at $A_{\text{body}} \approx 60°$, reaching about 1.07 BL/cycle—more than three times the best limbless viscous performance. Small amplitudes generate little displacement, and amplitudes above the peak do not continue to improve performance. The sharpness of the peak suggests strong coupling between body wave amplitude, leg contact timing, and frictional thrust generation: only a narrow range of amplitudes produces the right alignment between body bending and stance-foot geometry. This is consistent with experimental observations in centipede-like robots, where coordination between body and legs is known to be critical [1].

### 6.3 Phase Lag Sweep

**Procedure.** We fix the centipede body amplitude at 60° (the empirically optimal value identified in Sec. 6.2) and sweep the body–leg phase lag $\phi_{\text{lag}}$ from 0° to 360°. The performance metric is again displacement per cycle in BL.

**Interpretation.** Figure 4 shows that the centipede's locomotion performance depends strongly on the body–leg phase lag. The global maximum is near $\phi_{\text{lag}} = 180°$, where displacement reaches

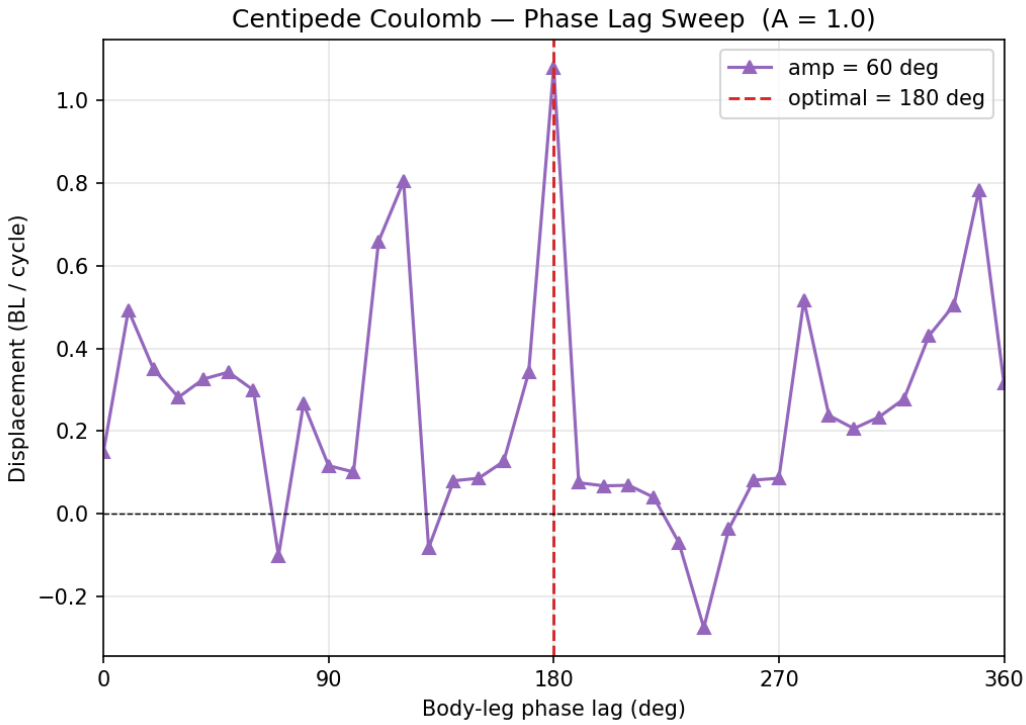


Figure 4: Centipede body–leg phase lag sweep at fixed body amplitude $A_{\text{body}} = 60°$. Performance is highly phase-dependent: the global maximum lies near $\phi_{\text{lag}} = 180°$ (red dashed line, ∼1.07 BL/cycle), with secondary local peaks elsewhere and several phase windows where the displacement is small or even negative. This shows that body amplitude alone does not determine performance; body–leg timing is equally critical.

about 1.07 BL/cycle. Several local peaks appear at other phase values (e.g., near 120° and 345°), but 180° is dominant in this simulator configuration. Equally informative are the phase windows where displacement is small or *negative*: poor coordination can reduce locomotion or even reverse it. This confirms that the centipede model is not controlled by body amplitude alone; the relative timing between the body wave and leg contact is critical for converting frictional contact into forward thrust.

### 6.4 Limbless Trajectories

Figure 5 shows $x(t)$, $y(t)$, and $xy$-trajectories for body amplitudes of 30°, 50°, and 70° in anisotropic viscous fluid. Forward displacement grows substantially from 30° to 50°, and remains close to that level at 70°, consistent with the saturation observed in Fig. 3. The lateral $y$-motion shows oscillations associated with the traveling wave, as well as a slow drift due to the asymmetric phase of the wave relative to the chosen body frame. The overall trajectories are smooth, consistent with the continuous nature of the viscous force model.

### 6.5 Centipede Trajectories

Figure 6 shows the same diagnostics for the centipede Coulomb system. The trajectories are visibly more irregular than the limbless viscous case. This is expected: Coulomb friction is non-smooth,

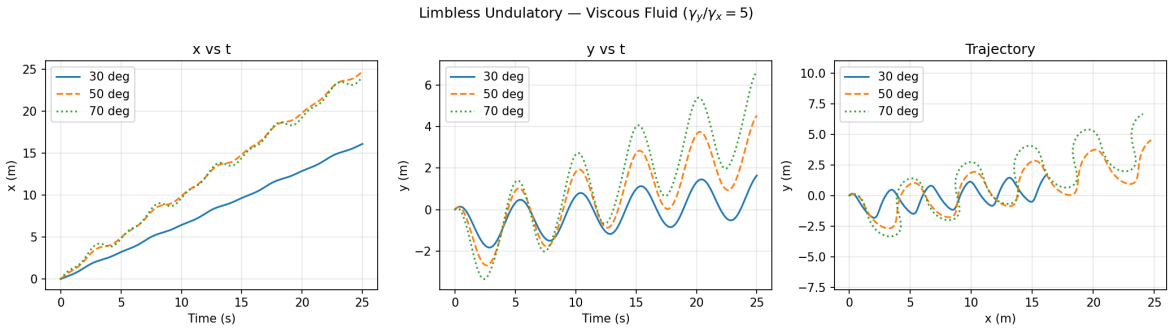


Figure 5: Trajectories of the limbless undulatory locomotor in anisotropic viscous fluid ($\gamma_y/\gamma_x = 5$) for body amplitudes of 30°, 50°, and 70°. Forward $x$-displacement (left) increases substantially from 30° to 50° and remains comparable at 70°, consistent with the saturation seen in Fig. 3 (left). The lateral motion (middle) grows with amplitude and the resulting paths (right) are smooth forward trajectories with growing lateral excursions, consistent with continuous viscous drag.

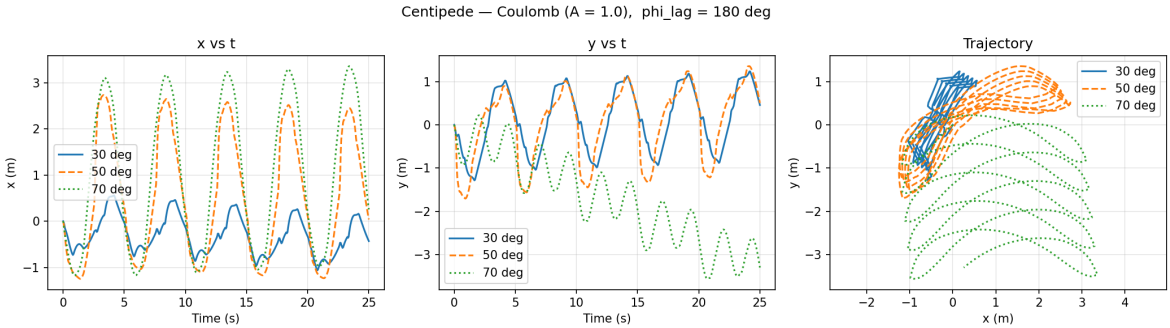


Figure 6: Trajectories of the centipede Coulomb system at body–leg phase lag $\phi_{\text{lag}} = 180°$ for body amplitudes of 30°, 50°, and 70°. Compared with the limbless viscous case, the trajectories are more irregular due to intermittent stance/swing transitions and Coulomb friction. Larger amplitudes generate larger oscillations and lateral drift, illustrating that high amplitude does not monotonically translate into better forward motion.

and stance/swing transitions introduce abrupt changes in which legs contribute force at any given instant. At 50° and 70°, larger body oscillations and lateral drift become apparent, showing that amplitude alone does not improve locomotion—a result consistent with the sharp peak in the amplitude sweep (Fig. 3, right).

### 6.6 Body Snapshots at $A = 60°$

Figure 7 compares time-lapse body shapes at the optimal amplitude for both systems. The limbless system progresses smoothly along $+x$, with steady-state undulation that is largely periodic in the body frame. The centipede system shows stronger shape reorientation and more complex spatial paths, consistent with the influence of leg contact and frictional thrust. These qualitative differences match expectations: a continuous viscous medium produces smoother motion than a non-smooth, intermittent contact model.

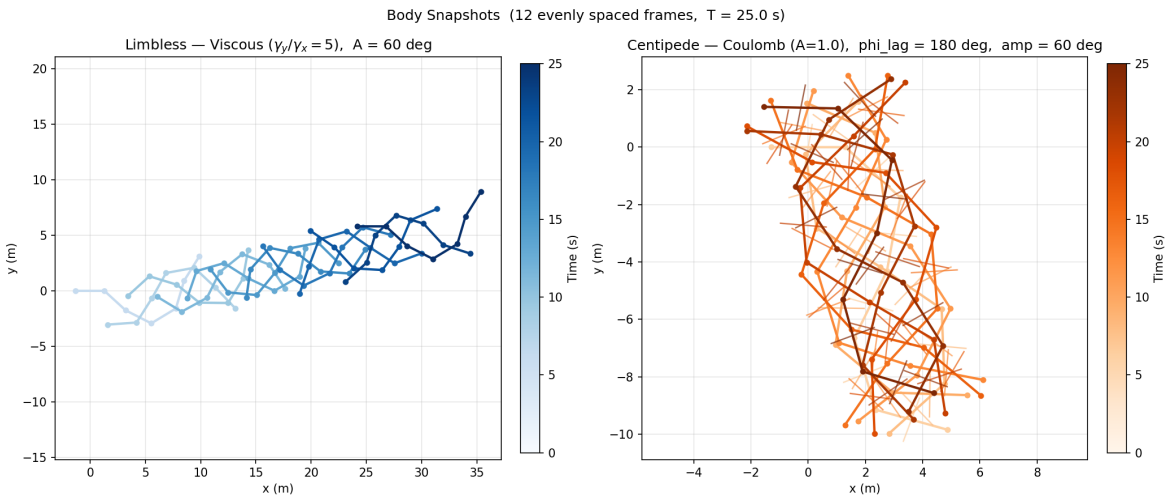


Figure 7: Body snapshots over $T = 25$ s (12 evenly spaced frames) at $A_{\text{body}} = 60°$. *Left:* limbless viscous case ($\gamma_y/\gamma_x = 5$); the body progresses smoothly along $+x$ with steady undulation. *Right:* centipede Coulomb case ($\phi_{\text{lag}} = 180°$); the body shows stronger reorientation and more complex paths due to intermittent leg contact and frictional thrust. The figure provides a qualitative confirmation that the simulator produces physically interpretable motion in both regimes.

### 6.7 Coulomb Nonlinearity and Solver Convergence

The amplitude and phase-lag sweeps in Figs. 3 (right) and 4 contain features that are not purely physical. In particular, the phase-lag sweep is visibly noisy, with displacement values that fluctuate sharply between adjacent phase points, and the amplitude curve shows a steep, narrow spike at $A_{\text{body}} = 60°$. A portion of this irregularity reflects the underlying mechanics of non-smooth contact, but a portion is numerical and deserves explicit discussion.

The Coulomb friction force law contains a sign function in the local velocity, which makes the local force law non-smooth in $\dot{g}$. As a consequence, the global force balance (Eq. 8) is no longer a smooth nonlinear system: the residual $W_{\text{net}}(\dot{g})$ has corners and small flat regions associated with stance/swing transitions and with link velocities that pass through zero. In addition, the contact schedule itself flips legs from stance to swing in discrete events, which introduces step changes in the structure of the equation being solved at each time step. The standard nonlinear solver used in `solver.py` is designed for smooth residuals, and near these transitions it can:

- converge to a local solution that is feasible but not the physically intended root,
- produce a body twist whose magnitude depends sensitively on the initial guess (the previous time step), and
- , in rare cases, fail to fully converge within the iteration budget, leaving a small but non-zero residual wrench.

In practice, this means that two simulations with very similar inputs (e.g. adjacent values of $\phi_{\text{lag}}$) can land on slightly different sequences of stance/swing decisions, which in turn produce slightly different per-step body twists, and—over many cycles—visibly different displacement-per-cycle values. This is the main reason the centipede phase-lag curve in Fig. 4 is jagged rather than smooth, and why the amplitude curve in Fig. 3 (right) shows a narrow rather than rounded peak.

The *trends* (strong sensitivity to phase, optimum near 180°, optimum amplitude near 60°) are robust, but the *point-wise* displacement values should not be over-interpreted.

A more careful treatment would replace the sign function with a smoothed approximation (e.g. a tanh regularization), enforce warm-starting between adjacent parameter values, and use a solver that is aware of complementarity at stance/swing transitions. These changes are listed as future work in the next section.

### 6.8 Validation Discussion

We do not claim full biological validation. Instead, we treat validation as a combination of three internal consistency checks:

**(1) Limiting cases.** Zero amplitude produces (essentially) zero displacement in both the viscous and the centipede systems (Fig. 3). Isotropic viscous drag yields no net displacement at any amplitude (Fig. 3, left), in agreement with the principle that anisotropy is required for propulsion in linear, drag-dominated swimmers.

**(2) Physical trends.** Anisotropic drag enables forward propulsion; intermediate amplitudes outperform very small ones; and centipede performance depends strongly on body–leg phase lag, with a clear optimum near 180°. These trends are consistent with the geometric-mechanics expectation that displacement is governed by enclosed area in shape space, and with the broader literature on coordinated body-leg locomotion.

**(3) Numerical consistency.** At every time step, the solver enforces the force-free condition (8); residual force/torque norms are monitored during integration. Trajectories and snapshots are visually inspected and produce physically interpretable motion.

**Confidence intervals.** The current results are deterministic parameter sweeps run on a fixed gait family, so we do not report confidence intervals. Fabricating intervals from a single deterministic run would be misleading. Future work could add stochastic perturbations (e.g., sensor noise, contact-coefficient variability, or random phase offsets) and use repeated trials to obtain proper confidence intervals on displacement per cycle.

## 7 Discussion, Conclusions, and Future Work

**Main findings.** The simulator reproduces the basic expectations of overdamped undulatory locomotion. Environmental anisotropy is necessary for limbless viscous propulsion: with $\gamma_y/\gamma_x = 1$ the body goes nowhere, and with $\gamma_y/\gamma_x = 5$ it makes forward progress. Gait amplitude has a non-monotonic effect on displacement in both the limbless viscous and centipede Coulomb systems, with optima near $A_{\text{body}} = 55$–$60°$. Centipede locomotion is strongly sensitive to body–leg phase lag, with a sharp peak at $\phi_{\text{lag}} = 180°$ and several phase windows where the body actually moves backward. The combined amplitude–phase analysis indicates that effective centipede locomotion requires both a sufficiently large body wave *and* the right body–leg coordination, and that one cannot substitute for the other. Importantly, all of these results emerge from the *same* simulator, applied to limbless and legged systems through different force-model choices and gait inputs, showing that the modular reduced-order modeling approach (rigid-link body plus RFT-style local forces plus quasi-static force balance) is general enough to support unified comparison across morphologies.

**Limitations.** Several simplifications limit the strength of the conclusions. The body is represented as a chain of rigid links rather than a continuous rod, which discretizes any traveling wave. The local force models are minimal: viscous drag is linear and isotropic per direction, and Coulomb friction is sign-dependent but otherwise constant. We do not perform an experimental calibration

against a specific animal or robot. The sweeps are deterministic, so we have no quantitative uncertainty bounds. Most importantly, the Coulomb friction force law is non-smooth, and—as discussed in Section 6.7—the resulting non-smooth global force balance can cause the nonlinear solver to converge to slightly different local solutions for nearby parameter values, especially near stance/swing transitions. This contributes visible numerical noise to the centipede phase-lag and amplitude sweeps, and means that point-wise displacement values should be interpreted with caution.

**Future work.** A natural next step is direct comparison with robot or animal data on body length per cycle as a function of body amplitude and body–leg phase lag. Adding parameter uncertainty (perturbed contact coefficients, jitter in leg phase) and running repeated trials would yield proper confidence intervals. The contact model could be improved with smoothed or regularized friction laws (e.g. a tanh approximation of the sign function), warm-started solver iterations between adjacent parameter values, and complementarity-aware stance/swing handling, which together would substantially reduce the numerical noise observed in Section 6.7. Gait optimization (e.g., over spatial wave number, body amplitude, and phase lag) using the existing simulator is a direct extension. Geometric height-function visualization would make gait design more intuitive by exposing which loops in shape space contribute to displacement. Finally, extensions to rough terrain or granular media (using the BB empirical force model already implemented) would push the framework toward realistic outdoor environments.

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# A Division of Labor

This project was completed by a single team member. **Jianfeng Lin** was responsible for the full project, including conceptual modeling, literature review, mathematical formulation, the Python implementation of all modules (`kinematics.py`, `forces.py`, `solver.py`, `simulation.py`, and the test scripts), all simulation experiments and parameter sweeps (amplitude sweep, phase-lag sweep, trajectory and snapshot studies), figure generation and visualization, and final report writing.

# B AI Use Statement

The authors acknowledge the use of ChatGPT (OpenAI) and Claude (Anthropic) for debugging, code refactoring, test development during programming, and language refinement in the final report.